Image Analysis Application
User Manual

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1 Purpose of the Application

This application is designed for **scientific analysis of image series**, typically originating from optical experiments such as laser beam diagnostics, imaging sensors, or laboratory camera systems.

The software allows users to:

- Load and process multiple images at once
- Automatically extract experimental parameters from file names
- Select regions of interest (ROI)
- Compute intensity, contrast, and profile metrics
- Analyze trends versus experimental variables
- Export numerical results, plots, and reports

No programming or scripting knowledge is required to operate the application.

2 Starting the Application

After launching the program, the main application window appears. The interface is divided into the following functional areas:

1. **Control panel** – loading images and selecting analysis options
2. **Image viewer** – displaying images and defining ROIs or cross-sections
3. **Results and plots area** – numerical output and trend plots
4. **Progress and log area** – monitoring batch processing

3 Loading Images

3.1 Supported Image Formats

The application supports the following image formats:

- JPG / JPEG
- PNG
- TIFF / TIF
- BMP

Both grayscale and RGB images are supported. The image type is detected automatically.

3.2 How to Load Images

1. Click the **Load Images** button
2. Select one or multiple image files
3. Confirm the selection

All selected images are treated as a single batch and processed together.

4 Automatic Parameter Detection from File Names

The application automatically extracts experimental parameters from image file names.

4.1 Example

A file name such as:

`laser_I=120mA_T=25C_exposure10ms.png`

may result in extracted parameters like:

- Current = 120
- Temperature = 25
- Exposure = 10

4.2 Purpose of Parameter Extraction

Extracted parameters are used to:

- Associate results with experimental conditions
- Generate trend plots (e.g. intensity vs. current)
- Export tables and reports with meaningful labels

Only numerical parameters are used for plotting and statistical analysis.

5 Viewing Images

After images are loaded:

- The first image is displayed automatically
- Images can be browsed or switched
- Correct scaling and color handling are applied

6 Region of Interest (ROI)

Often only part of an image is relevant for analysis, for example the illuminated beam region.

6.1 Available ROI Types

The application supports:

- Rectangular ROI
- Circular ROI
- Free polygon (user-defined shape)

6.2 How to Define an ROI

1. Select the desired ROI tool
2. Draw directly on the image
3. The ROI becomes active immediately

All calculations are performed only inside the selected ROI. The ROI can be cleared or redefined at any time.

7 Selecting the Analysis Type

The user can choose one of the following analysis modes.

7.1 Intensity Analysis

Computes:

- Total intensity (sum of pixel values)
- Squared intensity
- Participation ratio

This analysis is used to quantify energy distribution and beam uniformity.

7.2 Contrast Analysis

Computes several contrast measures, including:

- Michelson contrast
- RMS contrast
- Histogram-based contrast measures

This is useful for comparing image quality or uniformity.

7.3 Profile Analysis

Extracts intensity profiles along user-defined cross-sections. This is useful for studying beam shape, symmetry, and structure.

8 Defining Cross-Sections

For profile analysis:

1. Activate the cross-section tool
2. Draw one or more lines on the image
3. Each line defines a profile direction

The application computes and displays the intensity profile for each cross-section and saves the numerical data.

9 Running Batch Analysis

After selecting images, ROI, and analysis type, click **Run Analysis**.

9.1 Batch Processing

- All selected images are processed automatically
- A progress bar indicates current status
- The application remains responsive during processing

10 Viewing Numerical Results

After analysis:

- Numerical results are displayed in the output panel
- Each result is associated with a specific image
- Extracted parameters are included

11 Trend Plots

Trend plots visualize computed metrics versus experimental parameters.

11.1 Creating a Plot

1. Select an experimental parameter for the X-axis
2. Select a computed metric for the Y-axis
3. Click **Plot**

Only valid numerical values are available for plotting.

12 Exporting Results

12.1 Numerical Data Export

Results can be exported to TXT or CSV files containing:

- File names
- Extracted parameters
- Computed metrics

12.2 Plot Export

Any displayed plot can be exported as an image file (PNG).

12.3 PDF Report Export

A complete PDF report can be generated, including:

- File names
- Experimental parameters
- Numerical statistics
- Included plots

This report is suitable for documentation, sharing, and publications.

13 Typical Workflow Summary

1. Launch the application
2. Load image files
3. (Optional) Define a region of interest
4. Select analysis type
5. Run batch analysis
6. Inspect numerical results
7. Plot trends versus parameters
8. Export data and reports

14 Intended Users

This application is intended for:

- Researchers
- Scientists
- Engineers
- Students performing experimental image analysis